Geometry

ANSWER KEY

Section 1: Angle Relationships & Polygons

1 Two angles are complementary. If the measure of one angle is 14° more than three times the other, what is the measure of the smaller angle?

a. 19° ✓

b. 24°

c. 38°

d. 71°

Let the smaller angle be x. Complementary condition:
$x + (3x + 14) = 90 \Rightarrow 4x + 14 = 90 \Rightarrow 4x = 76 \Rightarrow x = 19^\circ$.

2 What is the sum of the interior angle measures of a convex hexagon?

a. 540°

b. 720° ✓

c. 900°

d. 1080°

Sum of interior angles $= (n - 2) \times 180^\circ$. For $n = 6$: $(6 - 2) \times 180^\circ = 4 \times 180^\circ = 720^\circ$.

3 In $\triangle ABC$, the exterior angle at vertex C measures 125°. If one remote interior angle measures 55°, what is the measure of the other remote interior angle?

a. 55°

b. 65°

c. 70° ✓

d. 75°

Exterior angle theorem: exterior angle equals the sum of two remote interior angles:
$55^\circ + x = 125^\circ \Rightarrow x = 70^\circ$.

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Section 2: Triangles, Similarity & Special Right Triangles

4 In a right triangle, the legs have lengths 9 and 12. What is the altitude drawn to the hypotenuse?

a. 6.0

b. 7.2 ✓

c. 7.5

d. 8.0

Hypotenuse $c = \sqrt{9^2 + 12^2} = \sqrt{81 + 144} = 15$. Altitude $h = \frac{ab}{c} = \frac{9 \times 12}{15} = \frac{108}{15} = 7.2$.

5 In an isosceles right triangle (45°-45°-90°), the hypotenuse is $14\sqrt{2}$ cm. What is the perimeter of the triangle?

a. $28 + 14\sqrt{2}$ **cm** ✓

b. $28\sqrt{2}$ cm

c. 42 cm

d. $14 + 14\sqrt{2}$ cm

Each leg is $\frac{14\sqrt{2}}{\sqrt{2}} = 14$ cm. Perimeter $= 14 + 14 + 14\sqrt{2} = 28 + 14\sqrt{2}$ cm.

6 Two similar triangles have perimeters of 24 cm and 36 cm. If the area of the smaller triangle is 32 cm^2, what is the area of the larger triangle?

a. 48 cm^2

b. 64 cm^2

c. 72 **cm**^{2} ✓

d. 96 cm^2

Scale factor of sides $= \frac{36}{24} = \frac{3}{2}$. Ratio of areas $= (\frac{3}{2})^2 = \frac{9}{4}$. Larger area $= 32 \times \frac{9}{4} = 72$ cm^2.

Section 3: Circles, Arcs & Tangents

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7 A chord of length 16 cm is at a distance of 6 cm from the center of a circle. What is the radius of the circle?

a. 8 cm

b. 10 **cm** ✓

c. 12 cm

d. 14 cm

The perpendicular from center bisects the chord into halves of 8 cm each. Using Pythagorean theorem: $r = \sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10$ cm.

8 What is the area of a circle whose circumference is 16π?

a. 16π

b. 32π

c. 64π ✓

d. 128π

$2\pi r = 16\pi \Rightarrow r = 8$. Area $= \pi r^2 = \pi(8^2) = 64\pi$.

9 A central angle of 45° intercepts an arc in a circle with radius 8. What is the length of the intercepted arc?

a. π

b. 2π ✓

c. 4π

d. 8π

Arc length $s = \frac{\theta}{360^\circ} \times 2\pi r = \frac{45}{360} \times 16\pi = \frac{1}{8} \times 16\pi = 2\pi$.

Section 4: 3D Surface Area & Volume

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10 What is the total volume of a right circular cylinder with radius 5 cm and height 12 cm?

a. 60π cm^3

b. 120π cm^3

c. 300π **cm**^{3} ✓

d. 600π cm^3

Volume $= \pi r^2 h = \pi(5^2)(12) = \pi(25)(12) = 300\pi$ cm^3.

11 A rectangular water tank is 4 meters long, 3 meters wide, and 2 meters deep. If it is currently filled to 75% of its capacity, how many cubic meters of water are in the tank?

a. 12

b. 16

c. 18 ✓

d. 24

Total capacity $= 4 \times 3 \times 2 = 24$ m^3. Filled to 75%: $0.75 \times 24 = 18$ m^3.

12 What is the surface area of a sphere with a radius of 6 cm?

a. 72π cm^2

b. 144π **cm**^{2} ✓

c. 288π cm^2

d. 576π cm^2

Surface area formula: $A = 4\pi r^2 = 4\pi(6^2) = 4\pi(36) = 144\pi$ cm^2.

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